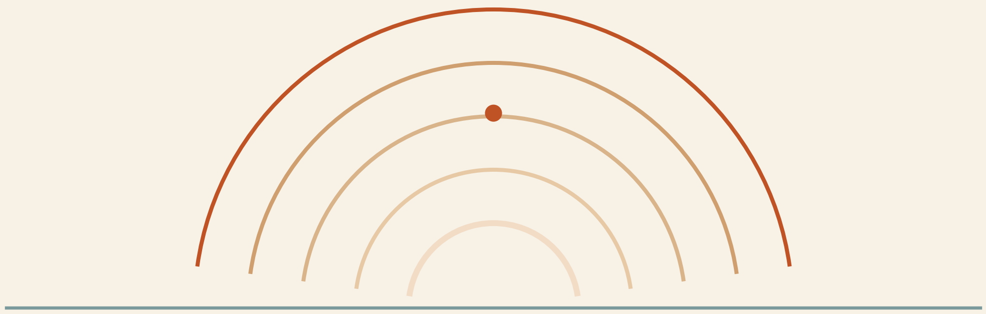
CONNECTED RESPONSE PLATFORM · FIELD REPORT

Kumbh Setu

One platform for pilgrims, volunteers and management — with **Kumbh Setu AI**, a voice-first field companion.



What this document covers

- What the platform is and how state flows between roles
- Access model — how you “become” each role (there is no login screen)
- The three dashboards + the Setu companion route, and what is in each
- Every Kumbh Setu AI feature, and how the ground-truth → Pulse → dispatch loop connects
- A step-by-step demo script for mentors, and an honest real-vs-simulated breakdown

SYNTHETIC / DEMO DATA THROUGHOUT

Generated for the Kumbh Setu prototype. Build: role apps + Setu AI Field Companion, ground intelligence, in-chat camera. Eval: 26/26.

1 What the platform is

Kumbh Setu is a connected response platform for a mass gathering. It has **three role experiences over one shared, live operational backbone**, plus **Kumbh Setu AI ("Setu")** — a voice-first AI field companion for volunteers. An action anywhere (a pilgrim SOS, a volunteer report, a control-room advisory) propagates to the other roles in real time.

It is a **prototype**. All data is synthetic. There is no server — state lives in the browser (Zustand store), syncs across tabs via `BroadcastChannel`, and survives reload via `localStorage`. "Reset simulation" in Management clears it.

2 Access model — there is no login screen

Authentication and role-based access control are deliberately deferred to a real pilot. For the prototype you **choose a role by opening its route**:

To act as…	Open	Identity switch
Pilgrim	/pilgrim	"Simulating a location in" dropdown (Ghat 1 … Pontoon Bridge)
Volunteer	/volunteer or /field	Volunteer dropdown top-right: V-101 … V-284 (each has a different zone, skills, languages)
Management / Control Room	/management	No switch — it is the operator view

So "log in as a volunteer" = open **/field**, pick **V-218** (M. Joshi, Ghat 4) from the dropdown. That is the demo default.

If a mentor asks about auth: role selection is by route today; production adds SSO + RBAC — Volunteer sees only their tasks and authorised zone data, Supervisor sees zone reports, Admin sees all. The tool layer already carries per-tool authorization checks, so RBAC slots straight into that.

3 The dashboards — how many, how to open

Three role dashboards, plus the Setu companion route and a scripted demo view. Run the dev server, then:

#	Surface	URL	For	Contains
1	Pilgrim app	/pilgrim	A pilgrim on a phone	Home (map, crowd band, advisories), SOS (one confirmed tap), Facilities, Lost & Found, Report an Issue (photo), Ask Kumbh Setu, offline-degradation sim
2	Volunteer app	/volunteer	A marshal in the field	Availability header; tabs: Task (accept → navigate → arrive → resolve/escalate, photo evidence), Team Chat (per-zone channel), Ask Setu (embedded companion)
2b	Kumbh Setu AI — Field Companion	/field	Same volunteer, Setu-first	Greeting + zone, tap-to-talk orb, live field status, quick actions, full companion (voice, translation, ground reports, camera). Bottom nav → Tasks / Control Room
3	Management control room	/management	Operators	Seven views (next table)
—	Landing	/	Product overview	—

#	Surface	URL	For	Contains
—	Scripted demo	/demo	Hands-off walkthrough	Runs the full SOS → resolution loop across a 3-pane synchronized view with paced delays; “Replay Scenario” resets and reruns

Management (/management) — seven views in the left nav:

View	What it shows
Live Operations	Operational map (abstract SVG or real Leaflet/OSM GIS of the Nashik–Trimbakeshwar corridor), active incidents, volunteer roster, top-risk zone Pulse panel
Kumbh Pulse	Explainable rule-based crowd-risk score per zone (0–100, contributor bars, model version + confidence) + Emerging Signals from aggregated field reports
Field Reports	Every ground report from Setu; corroborate, promote to incident, view photos and verification status
Advisories	Publish a zone or event-wide Notice / Advisory / Police Warning — appears live on Pilgrim apps
Lost & Found	Found-person reports auto-scored against open missing-person cases; one-click confirm match
Analytics	Response times, peak-activity histogram, category breakdown (illustrative baseline + live session data)
Event Log	Full audit trail — every material transition and every Setu action, timestamped

4 Kumbh Setu AI — feature list

Interaction

- Voice-first: tap-to-talk orb, live transcript, spoken replies (toggle), “Read aloud” replay, interrupt
- Uses the browser Web Speech API where available (Chrome desktop / Android); **always** has a text box + camera fallback for noisy conditions
- Visible states: idle / listening / thinking / searching / translating / taking action / waiting for confirmation / speaking / error / offline

Understanding

- 20-intent taxonomy (medical, lost person, crowd, water, sanitation, transport, ground report, my tasks, zone brief, translation, emergency, …)
- Short-lived session memory — resolves “she” / “there” / “the child” within a conversation; expires after 8 min idle or on “End”

Multilingual — English / हिन्दी (Hindi) / मराठी (Marathi) / தமிழ் (Tamil)

- One-shot: “translate ‘I need help’ into Tamil”
- Live translation mode: “help me talk to a Tamil-speaking pilgrim” → “tamil” → two-sided relay panel; each turn translated + playable aloud; confidence and phrasebook / best-effort shown
- LanguageService with detectLanguage / translate / speak / getSupportedLanguages

Grounded answers (RAG)

- Retrieves from a verified Kumbh knowledge base (SOPs, facilities, procedures) + live store data
- Shows the source line (“Source: Live facility database”, “Source: verified knowledge base”)
- Says “I don’t have verified information for that yet” instead of inventing — hallucination-resistant

Tool-calling — the model never touches state directly

- Read: find_nearest_facility, get_zone_status, get_my_tasks, get_resource_status, search_kumbh_knowledge, …
- Low-risk write: create_ground_report
- High-risk (always a confirm card): create_incident, escalate_incident, update_task_status (resolve / escalate)
- Every executed tool writes an audit event

Ground reporting

- “report a water shortage here” → Setu asks only what is missing (e.g. headcount) → structures category / severity / people affected / GPS / time / source / AI-confidence → you review → Submit
- Ground-truth ladder: unverified → reported → corroborated → verified → resolved; only an operator marks “verified”

Camera / image intelligence

- Camera button in the composer, anywhere in the chat
- Photo + a spoken note → vision provider returns a structured observation (label, category, potential impact, confidence) → ground-report draft with the photo attached → confirm → reaches Management
- Bare photo with no words → asks you to add a note; never guesses “fire” / “medical” unprompted

Safety & resilience

- Emergency mode — detects “collapsed / not breathing / stampede” → red UI, minimal questions, one confirm to create a CRITICAL incident + page nearest responder
- Never diagnoses; guides to verified medical resources; human-in-the-loop on every write
- Offline tolerance — ground reports queue locally and sync when back online; other writes are held

Engineering

- AIProvider / SpeechProvider / TranslationProvider / VisionProvider interfaces with working mock implementations — no API keys, no external calls
- A real model plugs into src/ai/providers/index.ts with no UI change
- docs/AI_FIELD_COMPANION.md (architecture, tools, safety, setup); npm run eval:setu → 26/26 scenarios

5 How the loop connects — the part that wins

Every arrow below is real state, synced live, and written to the audit trail.

Stage	What happens
1 Observe	Volunteer (Setu): “report water shortage, ~200 waiting” + photo → GR-40xx created (status: reported), audit event written
2 Corroborate	More volunteers report the same thing → Field Reports view; each “Log corroborating report” raises the count
3 Signal	Kumbh Pulse aggregates the reports (+ pilgrim requests + a resource flag) into one EMERGING SIGNAL with a confidence that rises as corroboration grows
4 Decide	Operator clicks “Promote to incident” → nearest qualified volunteer dispatched (skill + language aware, not just distance)
5 Resolve	Volunteer: Accept → Arrive → Resolve (resolve asks to confirm) → Management sees the full audit trail

The point: several weak field signals became one thing worth a human’s attention — and the AI never acted on its own.

6 How to run it

In the project folder (`C:\Users\ASUS\OneDrive\Desktop\KumbhSetu`):

Command	Purpose
npm run dev	Start the app. The terminal prints the URL — port 3000 is usually taken by another project on this machine, so it may say 3001 or a random port. Use whatever it prints.
npm run eval:setu	Run the 26 Setu reasoning-layer checks
npm run build	Production build (all 7 routes)

Open **/field** for the volunteer companion, **/management** for the control room. For the synchronized story, open **/pilgrim**, **/field** and **/management** as three tabs in the same browser — they stay in sync.

7 Demo script for mentors (~6–8 min)

Setup: one browser, three tabs — /field, /management, /pilgrim (optional) — side by side. On /management, left nav → **Reset simulation** for a clean slate.

Opening line: "Most assistants answer questions. Kumbh Setu AI helps the people serving the Kumbh understand, communicate, act and report — and turns what they see on the ground into operational intelligence."

Scene 1 — Setu knows you (/field)

Point out the greeting by name, the zone, the live field-status card.

Type "Nearest medical camp" → answer with name, distance, load, and **Source: Live facility database**.

Scene 2 — Language bridge

Type "help me talk to a pilgrim" → "marathi" → live translation panel opens.

In "Pilgrim speaks मराठी", type a Marathi sentence → Setu relays it to English with a confidence score and a Play button.

"A volunteer and a pilgrim who don't share a language, having a conversation."

Scene 3 — Emergency, minimal friction

Type "a pilgrim near Gate 4 has collapsed and is not responding" → UI goes red, Setu asks only what it needs, then one confirm → CRITICAL incident created and dispatched.

Switch to /management → Live Operations → the incident is already there with a volunteer assigned.

Scene 4 — Ground truth → signal (the differentiator)

Back on /field: "report a water shortage here" → "about 200 people waiting" → review card (Water, High, ~200, GPS, UNVERIFIED). Optionally tap the camera button and add a photo → Submit.

/management → Field Reports → the report is there. Click "Log corroborating report" 2–3 times.

/management → Kumbh Pulse → an Emerging Signal for water shortage now shows, confidence rising with corroboration.

Scene 5 — Close the loop

From the signal or Field Reports, "Promote to incident" → dispatch.

/field → "my tasks" → Accept → "mark arrived" → "mark this task resolved" (this one asks you to confirm).

/management → Event Log → scroll the audit trail: every step, every Setu action, timestamped.

Scene 6 — Honesty slide

Open the Kumbh Pulse view, read the "How to read this" panel aloud. Point at the SIMULATION MODE tags.

"Rule-based Stage 0 over synthetic data, decision support not a safety certification, and the AI runs on a local mock engine — the provider interface is built so a real model drops in without touching the UI."

Fallback if anything misbehaves live: open /demo — it runs the SOS → resolution loop automatically across three panes; “Replay Scenario” reruns it.

8 Real vs simulated (for mentor Q&A)

Genuinely functional	Simulated / demo
Full incident lifecycle state machine + skill/language-aware dispatch	All zones, facilities, volunteers, incidents, risk numbers
Real cross-tab sync (BroadcastChannel), localStorage persistence	Cross-device sync (needs a real backend)
Real audit trail from actual dispatches	Kumbh Pulse = rule-based Stage 0, not a validated model
Setu intent routing, tool policy, confirm gates, ground-report structuring, offline queue — all real logic	Setu LLM / speech / translation / vision = local mock providers (no keys, no calls); translation is a field phrasebook
Real Leaflet/OSM GIS map + 5 real registered hospitals/police posts (NTKMA open data, reference only)	Abstract SVG map geometry

9 Known limitations (say these first)

- No auth — role by route
- Mock AI providers — deterministic rule engine, phrasebook translation (real models plug in via the provider interface)
- Multilingual UI coverage is Pilgrim Home/SOS + the whole Setu companion, not every screen
- Cross-tab only, same browser
- Kumbh Pulse is explainable rule-based scoring, not a calibrated forecast
- Voice needs Chrome; everywhere else it is text + camera

Kumbh Setu — built for Kumbhathon Innovation S.P.R.I.N.T. 2026. This document and the platform use synthetic demo data throughout; nothing here represents real Nashik–Trimbakeshwar operational information.